IlmMate

Project Team

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Chapter 1

Introduction

This proposal aims to present the approach we will take with our project and provide a clear overview of the problem we are targeting, the motivation behind doing so, the proposed solution, and the scope of our work. With the increasing need for accessible education, especially in under-served areas, our focus is on creating a solution that empowers Pakistani students to learn at their own pace and convenience. IlmMate is a WhatsApp-based chatbot that will serve as an all-in-one educational tool, allowing students to access curriculum-specific explanations, fetch e-learning resources, and assess their progress through AI-generated quizzes. Through this project, we intend to tackle educational inequality in Pakistan while utilizing technology to solve the problem, offering students the tools they need to succeed. IlmMate aims to solve four of the UN's Sustainable Development Goals; 4 - Quality Education, 5 - Gender Inequality, 10 - Reduced Inequalities, and 17 - Partnership for the Goals.

1.1 Problem Statement

As a nation, Pakistan faces many challenges in global education, with one of the most alarming of these being its position as the country with the second-largest population of children out of school. [2] A staggering 37.7% of the population remains illiterate, which equates to approximately 60 million individuals [1] The root cause of this illiteracy and lack of education lies in insufficient educational infrastructure and barriers to education, particularly for females, differently-abled individuals, and the impoverished.

1.2 Scope

With IlmMate, we aim to develop a platform-independent educational chatbot that will uplift the learning experience of Pakistani grade 9 & 10 students. The chatbot's core purpose is to provide educational content with detailed explanations on various topics based on the curriculum from textbooks of Pakistani science subjects of grades 9 & 10. In this lieu, IlmMate will use an automated agent to provide users various e-learning resources, such as video links. It will also be able to generate questions based on topics selected by users, helping them practice these topics and reinforce their learning. IlmMate will generate reports on each student's performance. Lastly, for ease of use, the medium of communication will be WhatsApp.

1.3 Modules

Our product consists of three feature-oriented modules. Each module highlights specific features of IlmMate.

1.3.1 Backend Module: Content Management and Retrieval

In this module we handle the processing, storage and retrieval of content from vector database(s).

1. Digitization and Storage: We will use Optical Character Recognition (OCR) to digitize textbooks and store the data in vector databases for efficient retrieval.
2. Query Context Extraction & Similarly Matching: We will first shorten the user's prompt to just its core semantic meaning using context extraction techniques and then we will utilize algorithms to find and retrieve the most relevant content with respect to the shortened query from vector database(s).
3. Chatbot Response Generation: We will use a Large Language Model (LLM) to generate accurate and contextually correct responses using the previously received content from our vector database(s) as context.
4. Progress Reporting: IlmMate will track each student's performance through assessments and generate reports.

1.3.2 Client Module: WhatsApp Chatbot

This module deals with delivering specific features to users via WhatsApp.

1. Automated Assessment: IlmMate will generate questions by utilizing prompt engineering and a choice of a user selected topic to help them practice and enhance their learning.

2. e-Learning Resource Agent: IlmMate will deliver e-learning resources, such as video links, by using an agent to retrieve resources further aiding learning on a specific topic.

3. WhatsApp API: IlmMate will establish communication with users solely through the WhatsApp API.

1.3.3 Development Module: System Integration and Testing

This module focuses on integrating all components and ensure their proper functioning.

1. System Testing: We will conduct thorough testing to validate the system's functionality, performance and user experience.

2. Guardrails: We aim to set up safeguards to avoid responses for out of context queries and possibly inaccurate responses to user queries.

1.4 User Classes and Characteristics

This section outlines the user classes for IlmMate and how the project will affect them.

Table 1.1: User Classes and Their Descriptions

User class	Description
Student	Students are the main target users of IlmMate, specifically those in grades 9 and 10. They will converse with IlmMate to access curriculum-specific content, quizzes, and e-learning resources per their needs, allowing them to learn at their own pace & convenience.
Parent	Parents will receive regular reports on their child's performance. This will help them figure out areas where their child may need extra support, encouraging active involvement of parents.
Teacher	Teachers can be included to allow them to track progress of students enrolled in their classrooms through IlmMate. IlmMate could provide insights into student performance, helping teachers identify topics where students may be struggling and tailor their classroom lessons accordingly.
Educational Institution	Educational institutions can integrate IlmMate into their existing curricula to better the learning experience for pupils. Through IlmMate, institutions can provide supplementary support without needing additional resources for upskilling or upscaling faculty.

Example User Classes and Characteristics

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-Case Diagram

The use case diagram below describes the way that a user would interact with an use IlmMate.

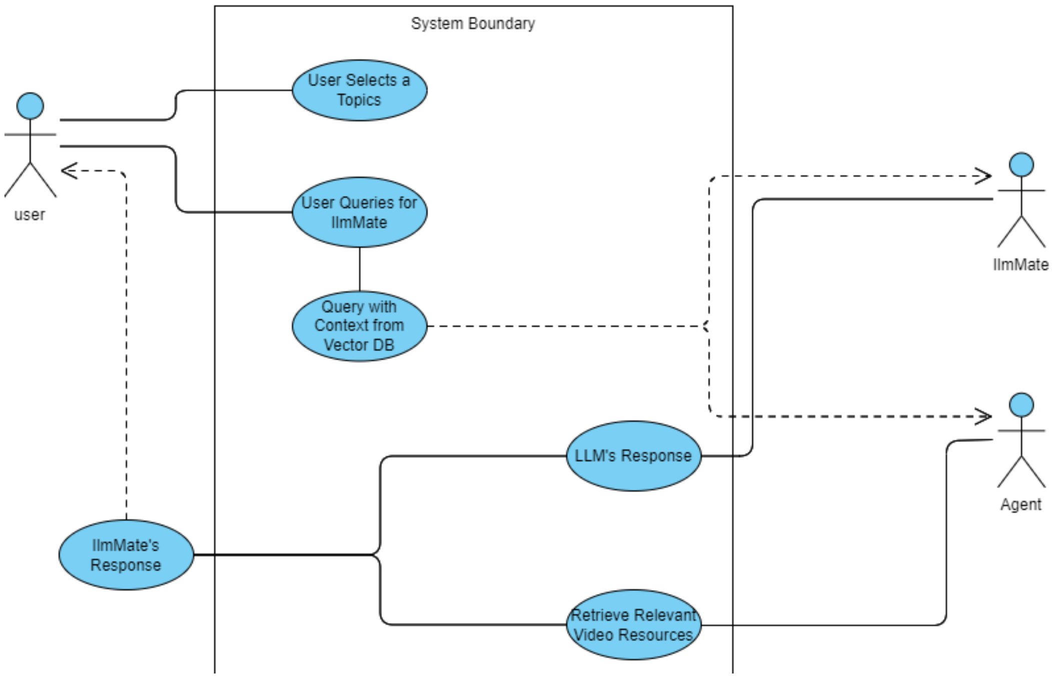


Figure 2.1: Activity Diagram for IlmMate System

2.2 Functional Requirements

The following are the functional requirements for IlmMate.

2.2.1 Backend Module: Content Management and Retrieval

Following are the Functional requirements for module 1:

1. The system should use OCR to convert textbooks into digital text and store it in a vector database.

2. The system must shorten user queries to focus on their main meaning.

3. The system should find and retrieve relevant content from the vector database

4. The system should track student performance and generate progress reports

2.2.2 WhatsApp Chatbot

Following are the requirements for module 2:

1. The system should communicate with users through the WhatsApp API.

2. The system should automatically generate practice questions based on topics chosen by users.

3. The system must provide e-learning resources, such as video links, based on user requests.

2.2.3 System Integration and Testing

Following are the requirements for module 3:

1. The system must integrate all parts of IlmMate

2. The system must have safeguards to manage off-topic queries and reduce incorrect responses.

3. The system must perform testing to ensure everything works correctly.

2.3 Non-Functional Requirements

This section outlines the non-functional requirements essential to the successful operation of IlmMate. These requirements focus on quality attributes, including reliability, usability, performance, and security, ensuring that the system meets user expectations and operates within defined parameters.

2.3.1 Reliability

IlmMate is designed with high reliability in mind to ensure minimal downtime and consistent operation. The system must achieve a Mean Time Between Failures (MTBF) of at least 48 hours, reflecting its ability to run continuously for extended periods without failure. In case of a failure, the system should implement automatic recovery mechanisms, sending alerts to developers and in turn minimizing user disruption. Regular maintenance and system monitoring will be employed to further enhance reliability and preemptively address potential issues.

2.3.2 Usability

Usability is a top priority for us, ensuring that all users can easily interact with the system, regardless of their technical proficiency. The chatbot interface, which is already extremely accessible through WhatsApp, will be intuitive and simple to navigate. Users should be able to send queries, request explanations, and receive responses without complex steps.

To enhance the user experience, the interface will provide clear feedback during the response generation process. This includes the use of filler sentences, which reassure users that their queries are being processed. Examples of these filler sentences include:

> "Let me get the answer to that for you."
> "Hold on a second while I think."
> "Just a moment while I look that up."
> "I'm fetching the information you need."
> "Give me a second to find the best answer."

These prompts will not only keep users engaged but also reduce the perceived waiting time, making interactions feel more seamless and conversational.

2.3.3 Performance

The performance of IlmMate is crucial to maintaining user satisfaction, especially during peak usage times. The system must respond to user queries within 20 seconds, even during high traffic, ensuring no significant delay in providing feedback. To maintain efficiency, the system will be optimized to support up to 50 concurrent users without experiencing performance degradation, such as slower response times or system crashes. While we aim for a 90% accuracy rate in LLM-generated responses, continuous model fine-tuning and updates in the RAG pipeline will be conducted to further improve this accuracy over time.

2.3.4 Security & Privacy

Given the sensitive nature of educational data and that children are target users for the application, security is paramount for IlmMate. Only authenticated users will have access to the system's features, with secure authentication mechanisms in place to prevent unauthorized access. Furthermore, if possible the system could keep parents in sync with their child's performances, usage and chat logs.

2.4 Domain Model

The domain model captures the key entities, relationships, and interactions within IlmMate. This model serves as a blueprint for the system, illustrating how data is structured and managed throughout the application. Entities such as Users, Queries, Course Material, and LLM Responses are represented to show the logical structure of the data and their relationships.

This model not only provides clarity on the system's internal workings but also forms the foundation for database design, ensuring that the system's data flow aligns with user needs and system functionality.

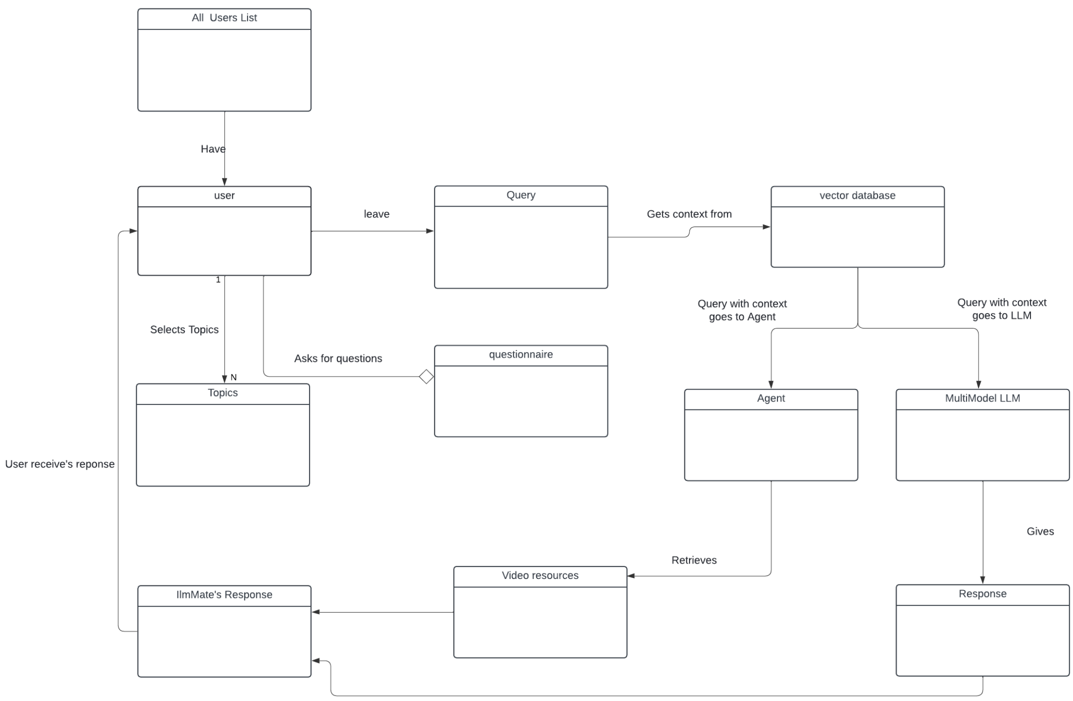


Figure 2.2: Domain Model Diagram for IlmMate

Chapter 3

System Overview

IlmMate is a platform-independent educational chatbot designed to enhance the learning experience for Pakistani grade 9 and 10 students. Its core functionality revolves around delivering curriculum-based educational content with detailed explanations, drawn from science textbooks. The chatbot leverages an automated agent to provide users with e-learning resources, such as video links, and generates topic-specific questions to help students practice and strengthen their understanding. Additionally, IlmMate tracks student progress and generates performance reports. The chatbot is designed to be easily accessible via WhatsApp for a seamless user experience.

3.1 Architectural Design

The architecture of IlmMate is designed with simplicity and effectiveness in mind. We've broken down the system into key components that work together to deliver the core functionalities. This high-level design allows us to focus on each part without getting overly complex, ensuring that the system remains scalable and adaptable as needed.

The structure enables us to handle tasks like processing user inputs, accessing and organizing learning materials, and generating assessments.

The initial design stage focuses on ensuring smooth data flow and system operations, with the potential to evolve into a more refined structure as the project progresses.

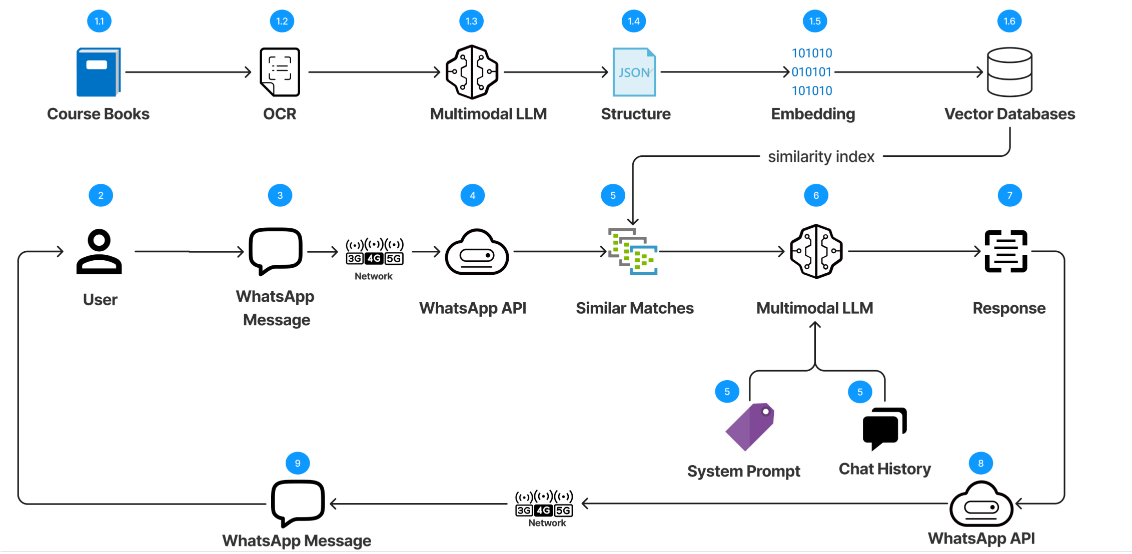


Figure 3.1: Architecture Diagram for IlmMate System

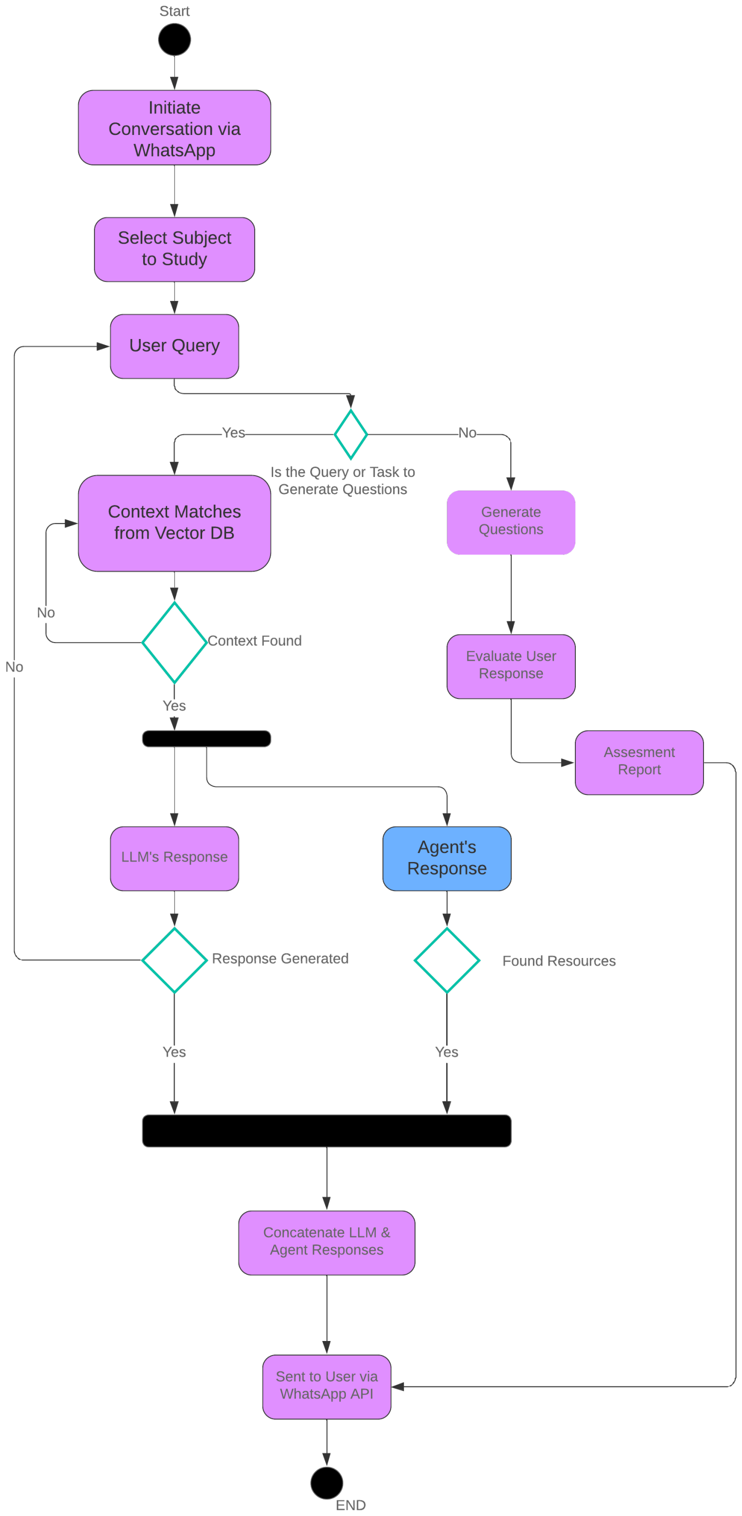
3.2 Design Models

The design models provide a structured representation of IlmMate's key processes, focusing on the system's interaction with data and users, and the sequence of events as the chatbot responds to user queries. This section includes activity, sequence, and data flow diagrams, illustrating how information flows through the system and how different components interact.

3.2.1 Activity Diagram

The activity diagram offers a high-level view of the operational flow within IlmMate, from when a user sends a query to when the chatbot processes the request and generates a response. It depicts how different processes are activated sequentially or in parallel to ensure smooth operation. The diagram visually outlines steps like message receipt, query processing, LLM interaction, and response generation, offering an easy-to-understand flow of the system's functionality.

Figure 3.2: Activity Diagram for IlmMate



3.2.2 System-Level Sequence Diagram

The sequence diagram demonstrates the order in which interactions occur within IlmMate, particularly how components such as the user interface, WhatsApp API, and the LLM work together to handle queries. This diagram is crucial for understanding how requests flow through the system—from the user input to processing and response delivery. By focusing on the sequence of method calls and interactions, this model highlights how modules exchange information in real time to generate educational content for users.

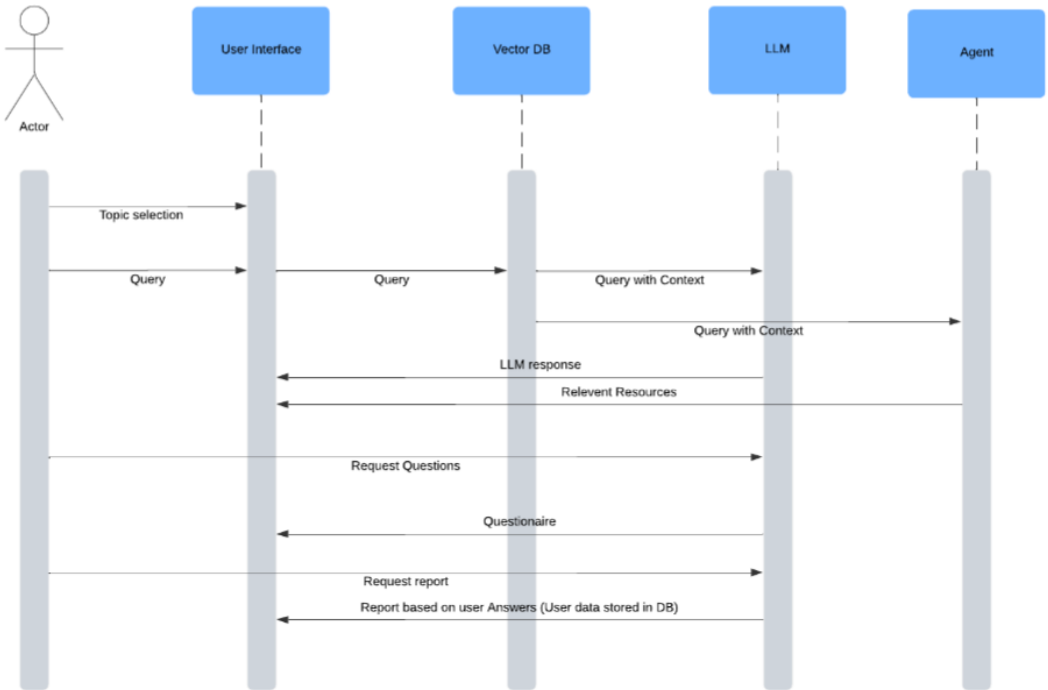


Figure 3.3: System-level Sequence Diagram for IlmMate

3.2.3 Data Flow Diagram

The data flow diagram (DFD) presents an overview of how data moves within IlmMate. It identifies key data sources (like user input via WhatsApp), data processing elements (LLM, database), and data sinks (response sent back to users). This diagram helps visualize the path that data follows as it travels through the system, beginning with the student's message, leading to the extraction of relevant information from textbooks or other learning resources, and concluding with a personalized response. The DFD emphasizes how data is structured, processed, and stored.

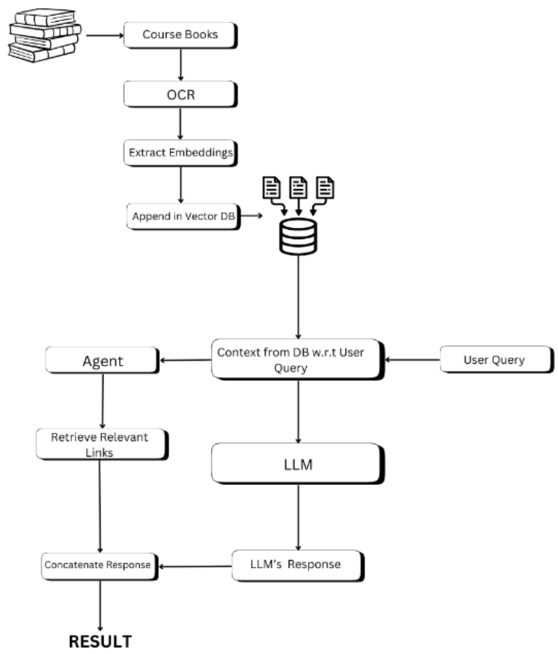


Figure 3.4: Data Flow Diagram for IlmMate

Design Models for Procedural Approach

The procedural approach in IlmMate's design simplifies each process into sequential steps that are easier to understand and implement. The key models for this approach include:

- **Activity Diagram:** Outlines the major operational steps, showing the flow from user interaction to LLM-based query handling.
- **Data Flow Diagram:** Breaks down the flow of information across multiple levels, detailing where data is coming from, how it's processed, and where it's sent.
- **System-level Sequence Diagram:** Describes the communication order between different parts of the system during its execution.
- **State Transition Diagram:** Highlights how IlmMate responds to various user queries and system events, tracking the states the system moves through from query receipt to response.

3.3 Data Design

IlmMate's data design revolves around efficiently processing and managing the educational content and student data required to deliver tailored learning experiences. The system's architecture ensures that data flows seamlessly between components, optimizing the chatbot's response accuracy and relevance.

3.3.1 Information Transformation

The information domain of IlmMate is structured around three main types of data: educational content, user inputs (student queries), and progress reports. These information types are processed through various stages in the system:

- **Educational Content:** IlmMate relies on textbooks and learning resources to provide detailed explanations and e-learning resources. These resources are initially scanned using Optical Character Recognition (OCR) and then transformed into digital text that can be structured and stored in vector databases as embeddings.
- **User Inputs:** The data from students (questions and requests) are processed through the WhatsApp API. These inputs are sent to the LLM (Large Language Model), which uses natural language processing and contextual understanding of the question - with reference to coursebooks - to generate a relevant response.
- **Progress Reports:** After generating AI-based assessments, student performance data is stored in a structured format and can be used to create personalized progress reports. These reports are generated by analyzing assessment results based on the curriculum topics the student has interacted with.

3.3.2 Data Storage and Organization

Data storage in IlmMate is designed to handle various types of information, with the following key components:

- **Vector Databases:** After the OCR process, embeddings of chunked textbook content is stored in a vector database. This database allows for efficient querying and similarity matching between the student's question and the stored content. Each textbook section or e-learning resource is embedded as a vector, facilitating faster retrieval of relevant information.
- **JSON Structure:** The system organizes input and output data (user queries and LLM responses) in JSON. This structure enables smooth communication between the model and the user. For example, when a student asks a question, their query is sent to the system as a JSON object, which then triggers the appropriate content retrieval and response generation.

- **User Data:** Information related to student performance, such as quiz scores and learning progress, is stored in a conventional database. This data is used to generate personalized progress reports that guide the student's learning journey. The system ensures that this data is securely stored and organized to maintain privacy.

IlmMate's data design is focused on ensuring that the right information is easily accessible, processed efficiently, and organized in a manner that supports smooth system functionality. By structuring the educational content, user inputs, and student performance data within the system, IlmMate is able to offer a personalized and adaptive learning experience for each student.

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